

UNIT 1-A: THE NERVOUS SYSTEM

Topic 3 Review: Sensory Reception

- 1) Determine whether each of the following statements are true or false. If a statement is false, correct it in the space provide(6)

_____ The purpose of sensory receptors is to signal effector cells to respond appropriately to changes in the internal or external environment.

_____ Sensory adaptation serves to filter out redundant sensory information in order to prevent overstimulation.

_____ The eye responds to changing intensity of light by adjusting the size of the lens, thereby controlling the amount of light that can enter the eye.

_____ The retina contains two types of photoreceptors: rods, which detect colour, and cones, which allow us to see in dim light.

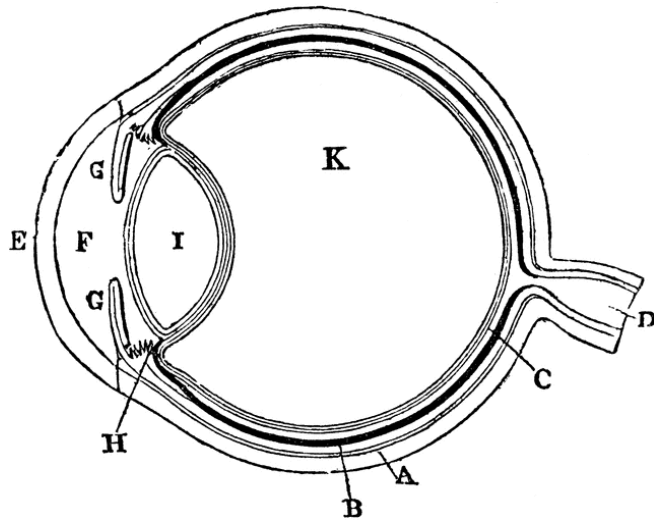
_____ Semicircular canals are fluid-filled structures within the ear that are essential for maintaining balance during movement.

_____ The oval window, also known as the “eardrum”, receives sound waves from the auditory canal and passes these vibrations on to the ossicles.

- 2) Label the following diagram of the eye(4)

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A = _____

B = _____

C = _____

D = _____

E = _____

F = Aqueous humour

G = _____

H = _____

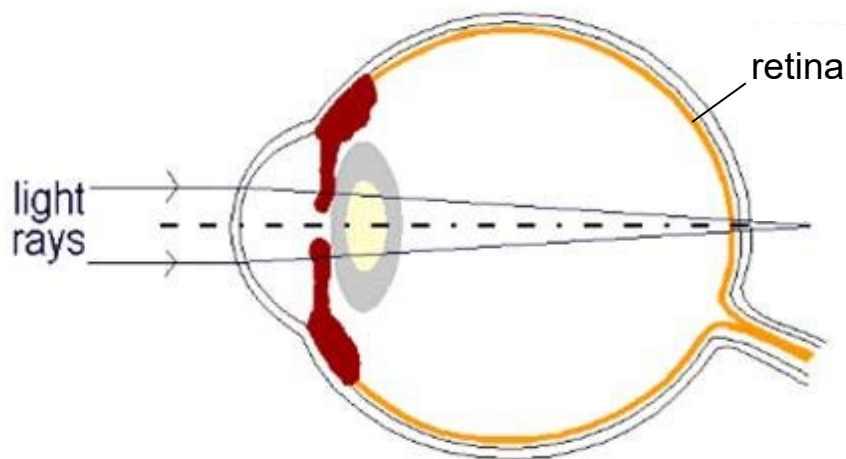
I = _____

- 3) Explain the difference between the role of the aqueous humour and the vitreous humour of the human eye(2)

- 4) Place the following structures in order as light passes through the eye(4):

Lens	Pupil	Cornea	Retina	Aqueous humour	Optic nerve
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5) Consider the following diagram of an abnormal human eye:



- Does the condition shown above represent myopia or hyperopia? Explain how you know. (2)
- Describe how this condition would affect a person's vision. (1)
- What type of lenses are required to correct this condition? (1)

6) Explain why a lack of vitamin A can lead to what is commonly referred to as "night blindness". (2)

- 7) Place each of the following statements in the correct order to describe how sound waves travel through the ear(5)

_____ Sound waves are directed by pinna to the auditory canal.

_____ An action potential is generated and travels along the auditory nerve to the brain to be processed & interpreted.

_____ Sound waves strike the tympanum, which generates vibrations within the middle ear.

_____ Vibrations are amplified by the ossicles and concentrated on the oval window, causing changes in pressure within organs of the inner ear.

_____ Changes in pressure stimulate the movement of hair cells within the cochlea.

- 8) Individuals who are frequently exposed to loud music often experience a deterioration in hearing over time; however, the extent of deafness differs depending on the frequency of the sound wave.

a. Which structure(s) in the ear have most likely become damaged in individuals with this type of hearing loss(1)

b. Explain why the loss of hearing is not equal for all frequencies(4)

Name: _____

TOTAL =

/26